Background

- In clinical trials, the participants are randomly allocated to either test drug or comparator group
- We will have different groups of subjects based on the number of treatments or treatment combinations being studied in the trial
- In parallel arm study, a subject will only receive the treatment to which he/she is randomized to
- To assess if the subject characteristics are similar across different treatment groups, we create summary reports
- Subject characteristics can be of numeric type (age, baseline BMI, etc) or categorical type(Sex, Race etc)
- For numeric or continuous variables, data is summarized in terms of sample size, number of records with missing data, mean, standard deviation, quartiles (q1, median(q2), q3), minimum and maximum etc
- These descriptive statistics are obtained for each group so that a comparison can be made between the groups
- Below is a table summarizing the age value and the results are presented in a tabular format with different treatments side by side for easy comparison
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Descriptive statistics for Age variable

Full Analysis Set

	Statistic	**Dose level 1**	**Dose level 2**	**Dose level 3**	**Total**
Age (years)	n (missing)	xx (xx)	xx (xx)	xx (xx)	xx (xx)
	Mean (SD)	xx.x (xx.xx)	xx.x (xx.xx)	xx.x (xx.xx)	xx.x (xx.xx)
	Median	xx.x	xx.x	xx.x	xx.x
	Q1, Q3	xx.x, xx.x	xx.x, xx.x	xx.x, xx.x	xx.x, xx.x
	Min, Max	xx, xx	xx, xx	xx, xx	xx, xx

Points to note in this kind of summary table

- The variable being analyzed is called an analysis variable.
- In the above example, age is the analysis variable
- The variable which contains the treatment group information is called a 'grouping' or 'class' or 'by' variable
- In the above example, we are creating a summary to compare different 'dose levels' - the variable that contains the 'dose level' of a subject will be called a 'class' variable
- If we follow ADaM standard, planned or actual treatment variables (TRT01P, TRT01A etc.) will be used as 'class' variables

- We need to check the different descriptive statistics that are being presented
- In this example, n, missing, mean, standard deviation, q1, median, q3, minimum and maximum are being presented
- The number of decimal values presented for different statistics are to be noted.
- Generally,
  - mean, median, q1, q3 will be presented to 1 additional decimal place than in data
  - minimum and maximum will be presented to the same number of decimals as in data
  - standard deviation will be presented to 2 additional decimal places than in data
- Some of the statistics are presented side by side
  - n (missing)
  - Mean (SD)
  - Q1, Q3
  - Min, Max
- Treatment groups are presented as columns in the table
- Statistics are presented as rows in the table with a specific order

How do we program this in SAS?

- There are multiple ways in which descriptive statistics for numeric variables can be obtained in SAS
- Most commonly used procedures for this purpose are proc means or proc summary
- Both proc means and proc summary can produce output to the output window or to an output dataset
- We need extensive transformations on the obtained numbers reported by these procedures
  - control the number of decimals to be presented for different statistics
  - concatenate two different statistics into a single row
  - use descriptive labels for the statistics etc
- We need to use a combination of data steps, proc steps (like sort, transpose, report etc) to produce the final output